

Figure 1: Block diagram of a database system

Now, let's have a look at why any application needs a database.

1. It will be difficult for any online app to store huge amounts of data for millions of its customers without a database.
2. Apart from storing data, a database makes it quite easy to update any specific data (out of a large volume of data already residing in the database) with newer data.
3. The data stored in a database of an app will be much more secure than if it's stored in any other form.
4. A database helps us easily identify any duplicate set of data present in it. It will be quite difficult to do this in any other data storage method.
5. There is the possibility of users entering incomplete sets of data, which can add to the problems of any application. All such cases can be easily identified by any database.

A user cannot directly interact with any database—there needs to be an interface or intermediate system, which helps the user to interact with it. Such an interface is referred to as a database management system (DBMS). It is basically a computer software application that interacts with the user or other applications, and even with the database itself, in order to capture and analyse the data. Any DBMS such as MySQL is designed in such a way that it allows the definition, querying, creation, updation and administration of the whole database. It is where we request the database to give us the required data in the query language.

Different types of databases

Relational database: This is one of the most common of all the different types of available databases. In such types of databases, the data is stored in the form of data tables. Each table has a unique key field, which is used to connect it to other tables. Therefore, all the tables are related to each other with the help of several key fields. These databases are used extensively in different industries and will be the type we are most likely to come across when working in IT.

Operational database: An operational database is quite important for organisations. It includes the personal

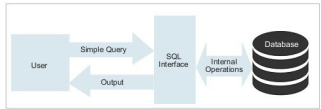


Figure 2: Retrieval of output data from a database using queries

database, customer database and inventory database, all of which cover details of how much of any product the company has, as well as the information on the customers who buy the products. The data stored in different operational databases can be changed and manipulated based on what the company requires.

Data warehouses: Many organisations are required to keep all relevant data for several years. This data is also important for analysing and comparing the present year data with that of the previous year, to determine key trends. All such data, collected over years, is stored in a large data warehouse. As the stored data has gone through different kinds of editing, screening and integration, it does not require any more editing or alteration.

Distributed databases: Many organisations have several office locations—regional offices, manufacturing plants, branch offices and a head office. Each of these workgroups may have their own set of databases, which together will form the main database of the company. This is known as a distributed database.

End user databases: There is a variety of data available at the workstation of all the end users of an organisation. Each workstation acts like a small database in itself, which includes data in presentations, spreadsheets, Word files, downloaded files and Notepad files.

Choosing the right database for your application

Choosing the right database for an application is actually a long-term decision, since making any changes at a later point can be difficult and even quite expensive. So, we cannot even afford to go wrong the first time. Let's see what benefits we will get if we choose the right database the first time itself.

1. Only if we choose the right database will the relevant and the required information get stored in the database, putting data in a consistent form.
2. It's always preferable that the database design is normalised. It helps to reduce data redundancy and even prevents duplication of data. This ultimately leads to reducing the size of the database.
3. If we choose the correct database, then the queries fired in order to fetch data will be simple and will get executed faster.
4. The overall performance of the application will be quite good.
5. Choosing the right database for an application also helps in easy maintenance.